



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
21-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 marks;
 choosing **ONLY** (B) will get +1 marks;
 choosing **ONLY** (D) will get +1 marks;
 choosing no option (i.e. the question is unanswered) will get 0 marks, and
 choosing any other combination of options will get -1 mark.

1. An infinitely long charged wire having linear charge density λ is placed along z-axis & a particle of mass m , charge Q is projected from $(R, 0, 0)$ with velocity v_0 along (+ve) y-axis in gravity free space Then.

(A) Speed of particle when its distance from wire grows to $2R$ is $\sqrt{v_0^2 + \frac{\lambda Q}{\pi \epsilon_0 m} \ln 2}$

(B) Speed of particle when its distance from wire grows to $2R$ is $\sqrt{v_0^2 + \frac{\lambda Q}{2\pi \epsilon_0 m} \ln 2}$

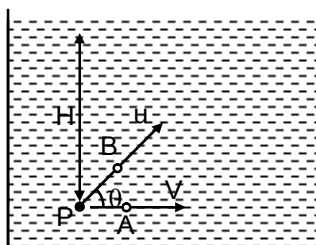
(C) Velocity component of particle along the radial line (joining the line charge to particle) at an instant its distance becomes $2R$ is $\sqrt{\frac{v_0^2}{4} + \frac{\lambda Q}{2\pi \epsilon_0 m} \ln 2}$

(D) Velocity component of particle along the radial line (joining the line charge to particle) at an instant its distance becomes $2R$ is $\sqrt{\frac{3v_0^2}{4} + \frac{\lambda Q}{\pi \epsilon_0 m} \ln 2}$

Space for Rough Work

2. Two small balls A and B are projected simultaneously from point P inside liquid as shown. A is projected horizontally with speed V and B at an angle θ to the horizontal with speed u . Liquid has density ρ , ball A has density $\frac{\rho}{2}$ and ball B has density ρ .

$H = 10\text{m}$, $\theta = 45^\circ$, $g = 10\text{ m/s}^2$. If the balls collide at the surface of liquid, then



- (A) $u = 10\sqrt{2}\text{ m/s}$ (B) $u = 10\text{ m/s}$
 (C) $u = 5\text{ m/s}$ (D) $V = 5\sqrt{2}\text{ m/s}$

Space for Rough Work

3. A spherical ball of mass m and radius r is projected from origin with an initial velocity $\vec{v} = v_0 \hat{i}$ in a viscous fluid having coefficient of viscosity η as shown. The space is gravity free but the ball is always acted by a force $\vec{F} = \vec{A} \times \vec{v}$ where $\vec{A}$ is a constant vector along positive z -axis and $\vec{v}$ is instantaneous velocity vector of the ball.

(A) The displacement of ball until the ball stops is $\frac{2mv_0}{\sqrt{A^2 + (6\pi\eta r)^2}}$

(B) The displacement of ball until the ball stops is $\frac{mv_0}{\sqrt{A^2 + (6\pi\eta r)^2}}$

(C) Motion of ball is always in x - y plane

(D) Speed of particle is $\left(\frac{v_0}{2}\right)$ after $\frac{\ln 2}{6\pi\eta mr}$ seconds.

4. The density ρ inside a solid sphere of radius R & mass M varies as $\rho \propto \frac{1}{r}$, where r is the distance from the centre. Then,

(A) Gravitational potential at the centre of sphere is $-\frac{2GM}{R}$

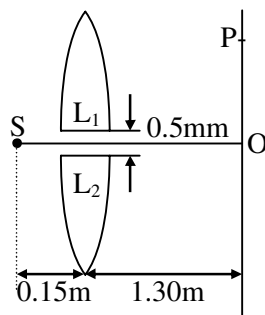
(B) Gravitational potential at the centre of sphere is $-\frac{GM}{2R}$

(C) Internal potential energy (self energy) of the matter forming sphere is $-\frac{4}{3} \frac{GM^2}{R}$

(D) Internal potential energy (self energy) of the matter forming sphere is $-\frac{2}{3} \frac{GM^2}{R}$

Space for Rough Work

5. Billet's split lens is a device consisting of two halves of convex lens placed close together to form two real or virtual images S_1 & S_2 of narrow slit S illuminated by a monochromatic source of light. In the shown figure, S is a point source emitting light of wavelength 500 nm and a thin convex lens of focal length 0.1 m is cut into identical halves L_1 and L_2 and are placed symmetrically about the central axis with a gap of 0.5 mm. The distance along the axis from S to L_1 and L_2 is 0.15 m, while that from L_1 and L_2 to O is 1.3 m. Choose the correct option(s) :



- (A) If third intensity maximum occurs at point P , distance OP will be 1 mm.
- (B) If third intensity maximum occurs at point P , distance OP will be 1.5 mm.
- (C) Reducing the original gap of 0.5 mm between L_1 and L_2 will increase the fringe width observed on screen.
- (D) Changing gap length between L_1 and L_2 will have no effect on fringe width observed on screen

Space for Rough Work

6. A rectangle of perfectly conducting wire having sides a and b ($b > a$), mass M , self inductance L , moves with an initial velocity v_0 in its plane, directed along its longer side, from a region of zero magnetic field into a region with a field B_0 which is uniform and perpendicular to the plane of rectangle. At $t = 0$, rectangle just starts to enter magnetic field. Choose correct statement(s)

(A) Velocity of rectangle as a function of time is given by $v = v_0 \cos \omega t$, where $\omega = \frac{B_0 a}{\sqrt{mL}}$

(B) Velocity of rectangle as a function of time is given by $v = v_0 (1 + \sin \omega t)$, where $\omega = \frac{B_0 a}{\sqrt{mL}}$

(C) Displacement of rectangle at any time t is $s = \frac{v_0 \sqrt{mL}}{B_0 a} \sin \omega t$, where $\omega = \frac{B_0 a}{\sqrt{mL}}$

(D) Displacement of rectangle at any time t is $s = \frac{v_0 \sqrt{mL}}{B_0 a} (1 + \sin \omega t)$, where $\omega = \frac{B_0 a}{\sqrt{mL}}$

Space for Rough Work

7. Disc A of mass m , radius r and disc B of mass $2m$, radius r are placed on horizontal smooth floor, Disc A is given angular velocity ω about its stationary vertical central axis and disc B is projected with horizontal velocity v without rotation for a head on collision to occur between them. If collision is perfectly inelastic and slipping between disc ceases in the end of collision, then after collision,

(A) Angular speed of disc A is $\frac{5\omega}{9}$

(B) Angular speed of disc B is $\frac{4\omega}{9}$

(C) speed of centre of disc B is $\frac{3v}{2}$

(D) speed of center of disc A is $\sqrt{\left(\frac{3v}{2}\right)^2 + \left(\frac{2\omega r}{9}\right)^2}$

8. Consider white light whose wavelength spread is from 400 nm to 700 nm. Its energy is uniformly distributed in this spectrum. The light is incident on a metal A of work function 1.55eV. Saturation photo current is 6mA. Now the same light is incident on metal B, work function 2.48eV. [Take $hc = 1240$ eV nm. Assume photo efficiency remains same]

(A) Stopping potential for experiment with metal A is 0.22 V

(B) Stopping potential for experiment with metal B is 0.62 V

(C) Saturation photo current for metal B will be 2mA

(D) Saturation photo current for metal B will be less than 2mA

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **Two (02)** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Eight** entries (P), (Q), (R), (S), (T), (U), (V) and (W).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

MTC-1

An adiabatic container containing ideal gas ($\gamma = 1.5$) is divided into 3 sections A, B and C with the help of pistons I and II of which piston II is non-conducting. Initially pressure in all the sections is P_0 . Leftmost part of container is slowly given heat through a heater so that final volume of section C

becomes V' . (Use $R = \frac{25}{3} \text{ J / mol - K}$)

The list-I contains some quantities involved in thermodynamic operation & list -II gives some possible values of these quantities.

List-I

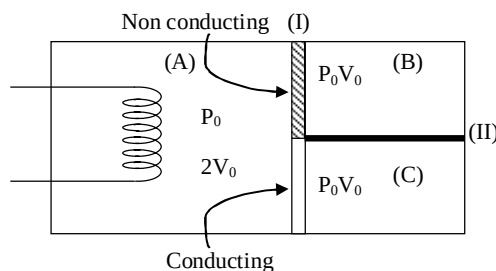
- (i) Final temperature in section (A)
- (ii) Final pressure in section (B) numerically is closest to
- (iii) Final pressure in section (A) numerically is closest to
- (iv) heat supplied by heater

List-II

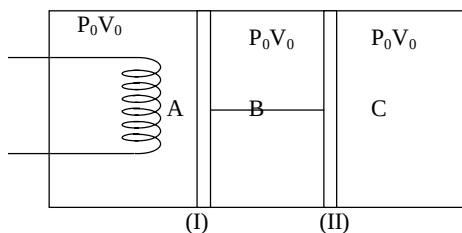
- (P) $3P_0$
- (Q) $18 P_0 V_0$
- (R) $\left(\frac{368}{9}\right) P_0 V_0$
- (S) $5 P_0$
- (T) $8 P_0$
- (U) $\frac{41}{45} P_0 V_0$
- (V) $\frac{63}{100} P_0 V_0$
- (W) $15 P_0 V_0$

Space for Rough Work

9. Piston (I) is fixed with lower half conducting and upper half non-conducting. Piston (II) is freely moving and massless. Section (A) has two moles, (B) and (C) each has 1 mole. If $V' = \frac{41V_0}{25}$, the correct match is :



- (A) (i) $\rightarrow$ (U), (ii) $\rightarrow$ (P), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (Q)
 (B) (i) $\rightarrow$ (U), (ii) $\rightarrow$ (S), (iii) $\rightarrow$ (T), (iv) $\rightarrow$ (R)
 (C) (i) $\rightarrow$ (V), (ii) $\rightarrow$ (P), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (R)
 (D) (i) $\rightarrow$ (U), (ii) $\rightarrow$ (S), (iii) $\rightarrow$ (T), (iv) $\rightarrow$ (V)
10. Piston (I) is conducting and pistons (I) and (II) are connected by a thin non-conducting rigid rod. Pistons can move freely. Section (A), (B), (C) each has 1 mole. If $V' = \frac{4V_0}{9}$, then the correct match is : [Note : section (B) is entire middle section]



- (A) (i) $\rightarrow$ (Q), (ii) $\rightarrow$ (S), (iii) $\rightarrow$ (P), (iv) $\rightarrow$ (W)
 (B) (i) $\rightarrow$ (Q), (ii) $\rightarrow$ (P), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (W)
 (C) (i) $\rightarrow$ (V), (ii) $\rightarrow$ (S), (iii) $\rightarrow$ (P), (iv) $\rightarrow$ (Q)
 (D) (i) $\rightarrow$ (V), (ii) $\rightarrow$ (S), (iii) $\rightarrow$ (P), (iv) $\rightarrow$ (W)

Space for Rough Work

MTC-2

In DC circuits, behaviour of inductor and capacitor is peculiar. When steady state is established in circuits, no current passes through capacitor and no potential difference appears across inductor. Current in branch AB is represented by z .

List (I) contains some quantities involved in circuits and list (II) gives some possible values of these quantities.

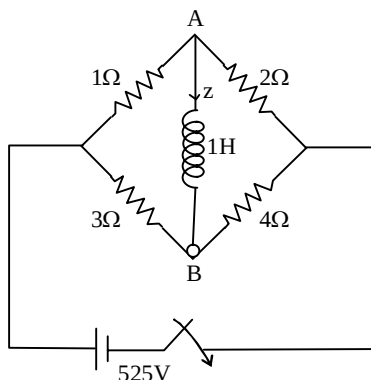
List-I

- (i) Current through battery immediately after switch is closed
- (ii) Magnitude of z long time after switch is closed
- (iii) Value of z as a function of time
- (iv) Current through battery long time after switch is closed

List-II

- (P) $\frac{1000}{3} \left(e^{-\frac{3}{2}t} - \frac{1}{4} \right)$
- (Q) $\frac{1000}{3} \left(e^{-21t} - \frac{1}{4} \right)$
- (R) 250A
- (S) $21 \left(1 - e^{-\frac{50t}{21}} \right) \text{A}$
- (T) 21A
- (U) 252A
- (V) $\frac{250}{3} \left(1 - e^{-\frac{50t}{21}} \right)$
- (W) $\frac{250}{3} \text{A}$

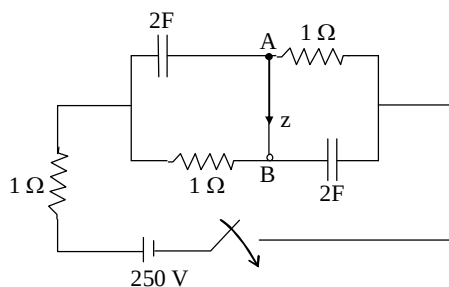
11. Switch is closed at $t = 0$.



- (A) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (T), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (R)
- (B) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (W), (iii) $\rightarrow$ (V), (iv) $\rightarrow$ (U)
- (C) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (T), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (U)
- (D) (i) $\rightarrow$ (U), (ii) $\rightarrow$ (T), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (R)

Space for Rough Work

12. Switch is closed at $t = 0$



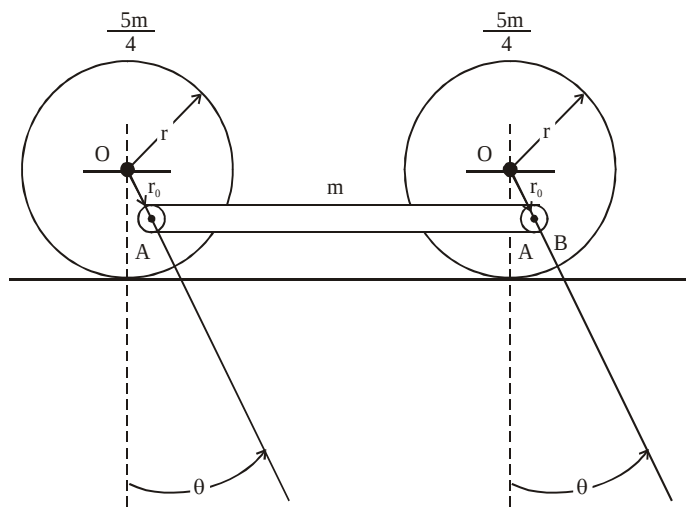
- (A) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (W), (iii) $\rightarrow$ (P), (iv) $\rightarrow$ (W)
 (B) (ii) $\rightarrow$ (R), (ii) $\rightarrow$ (W), (iii) $\rightarrow$ (Q), (iv) $\rightarrow$ (W)
 (C) (i) $\rightarrow$ (R), (iii) $\rightarrow$ (W), (iii) $\rightarrow$ (V), (iv) $\rightarrow$ (W)
 (D) (iv) $\rightarrow$ (W), (iv) $\rightarrow$ (R), (iv) $\rightarrow$ (Q), (iv) $\rightarrow$ (W)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

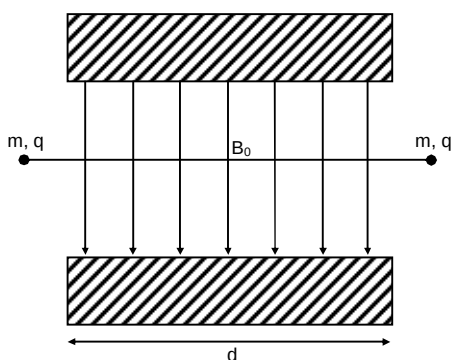
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. Two homogeneous solid cylinders each of mass $\frac{5m}{4}$ and radius r are connected by a thin connecting link AB of mass m as shown. If system is slightly displaced from its equilibrium, frequency of oscillation assuming no slipping on ground is found to be $\frac{1}{2\pi} \sqrt{\frac{g}{xr}}$. Find x ? (Point A and point B of link AB are at distance $r_0 = \frac{r}{2}$ from centre of their respective cylinders)

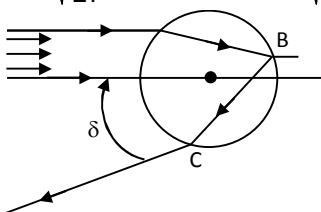


Space for Rough Work

2. Thin insulating weight less rod of length L may be freely rotated horizontally about a vertical axis passing through its middle. At the ends of the rod are fixed two small balls with mass m and charge q . This whole system is placed between the cylindrical poles of an electromagnet, creating a uniform vertical magnetic field induction B_0 . The diameter of the pole is equal to $d < L$, and their axis coincides with the axis of rotation of the bar (see figure). The magnetic field is uniformly reduced to zero. The angular velocity, acquired by the rod after turning off the magnetic field is $\left(\frac{x}{y}\right) \left(\frac{qB_0 d^2}{mL^2}\right)$. Find $x + y$? ($\frac{x}{y}$ in lowest form)

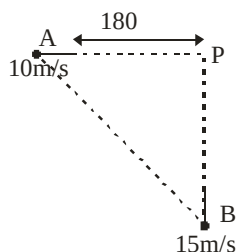


3. A parallel beam of light is incident on a spherical drop of water ($\mu = 4/3$). Consider refraction of light at air-water interface, then reflection at water-air boundary (of course there will be refracted light energy in air as well), followed by refraction at water-air interface. Path of a typical ray refracted at A, then reflected at B and finally refracted at C has been shown in the Figure. Find the maximum value of angle δ . (in degree) (use $\sin^{-1} \sqrt{\frac{20}{27}} = 60^\circ$ and $\sin^{-1} \sqrt{\frac{5}{12}} = 40^\circ$).



Space for Rough Work

4. Rate of evaporation (volume evaporated per unit time) of water kept in an earthen pot is proportional to the volume of water present in the pot. It is observed that it takes 48 hrs for 75% of the water kept in the pot to evaporate. The empty pot was placed below a tap from which water leaks in small drops each of volume v_0 . The drops fall at a uniform rate of n drops per hour. The volume of water in the pot 24 after it was kept below the tap is $\left(\frac{nv_0}{\ln 2}\right)x$. Find x ?
5. Two cars A and B are approaching a crossroad junction P as shown. At an instant when A is 180 m and B is 255 m away from P, A momentarily blows a horn of frequency 2.94 Hz. Find corresponding frequency observed by B? [$V_{\text{sound}} = 300 \text{ m/s}$]



6. A particle is projected with speed u from a point at a height h above ground such that maximum range on the ground is R . If $u^2 = 2gh$, find value of $\frac{10R}{h}$? (Take $\sqrt{2} = 1.414$)

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then

choosing **ONLY** (A), (B) and (D) will get +4 marks;

choosing **ONLY** (A) and (B) will get +2 marks;

choosing **ONLY** (A) and (D) will get +2 marks;

choosing **ONLY** (B) and (D) will get +2 marks;

choosing **ONLY** (A) will get +1 marks;

choosing **ONLY** (B) will get +1 marks;

choosing **ONLY** (D) will get +1 marks;

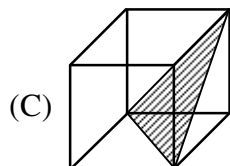
choosing no option (i.e. the question is unanswered) will get 0 marks, and

choosing any other combination of options will get -1 mark.

1. Select correct statement(s) :

(A) Zinc blende structure can also be imagined as an fcc arrangement of Zn^{2+} ions and S^{2-} ions occupying alternate tetrahedral voids.

(B) Fluorite structure has r_+/r_- less than 1. So, it can be also imagined as F^- ions forming simple cubic lattice and Ca^{2+} ions occupying alternate cubical voids.



(C) Shaded plane in a fcc lattice passes through centers of only one tetrahedral void and no octahedral void.

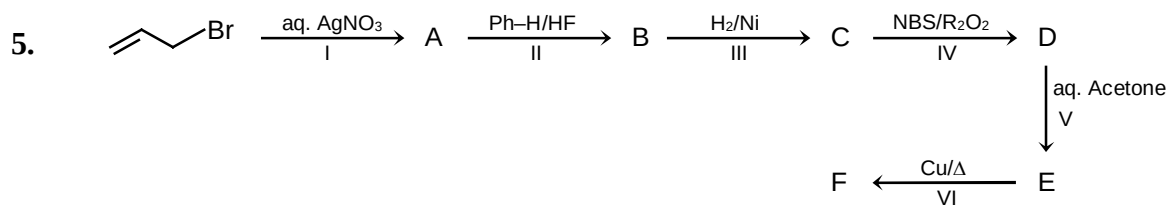
(D) If all A atoms form fcc lattice, B atoms occupy all tetrahedral voids and C atoms occupy only octa voids at body-centres of unit cell, then simplest formula of compound is $\text{A}_4\text{B}_8\text{C}$.

Space for Rough Work

2. Which of the following is/are chiral ?

- (A) mer-[Co(NO₂)₃(dien)] (B) [Ru(bipy)₃]⁴⁺
 (C) fac-[Co(NO₂)₃(dien)] (D) [Cr(ox)₃]³⁻
3. A system of nitrogen chemistry exists with many analogies to oxygen system based on water. Nitrogen forms amides, imides and nitriles from ammonia analogous to hydroxides and oxides derived from water. Thus, barium amide on thermal decomposition would form
- (A) Dinitrogen (B) Barium Nitride
 (C) Ammonia (D) Barium imide
4. Mixing which of the following combinations produce at least one gaseous product?
- (A) Ammonium nitrate solution and slaked lime
 (B) Copper metal and 0.10 M hydrochloric acid
 (C) Hypo solution and 0.1M hydrochloric acid
 (D) Zinc metal and alkaline solution of aluminium nitrate

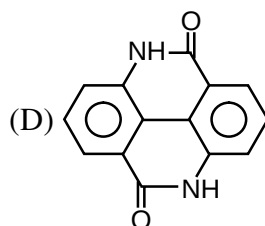
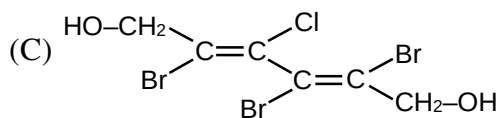
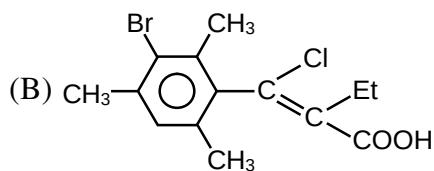
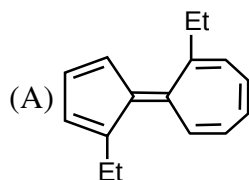
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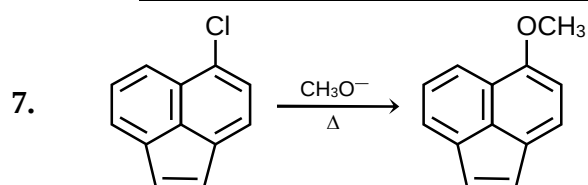
Which of the following statements is/are correct for the above reaction sequence ?

- (A) F gives yellow ppt with NaOI.
- (B) C gives total five monochloroproducts on photochemical chlorination.
- (C) E gives instant turbidity with Lucas reagent.
- (D) Step II involves electrophilic substitution reaction whereas step IV involves free radical addition reaction.

6. Which of the following compound is/are resolvable ?



Space for Rough Work



Which of the following statements is/are correct about above reaction?

- (A) This reaction is not possible as no electron withdrawing group is present on ortho and/or para to leaving group $-\text{Cl}$.
- (B) The nucleophile CH_3O^- is not strong enough to deprotonate the aromatic ring so benzyne mechanism is ruled out.
- (C) $\text{S}_{\text{N}}1$ like reaction is not possible because $-\text{Cl}$ is not good leaving group due to double bond character between carbon and chlorine.
- (D) This reaction takes place by $\text{ArS}_{\text{N}}2$ mechanism as Meisenheimer complex renders five membered aromatic ring as in cyclopentadiene.
8. Which of the following statement(s) is(are) true?
- (A) Monosaccharides generally can not show mutarotation.
- (B) Reduction of D-fructose with H_2/Ni gives only D-sorbitol.
- (C) Methyl α -D-glucopyranoside is non-reducing in nature.
- (D) Fructose and glucose can be distinguished by Seliwanoff's method.

Space for Rough Work

SECTION-I(ii): (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **Two (02)** Multiple Choice Questions.
- Each List-Match set has two lists: **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

Answer Q.9 and Q.10 by appropriately matching the lists based on the information given in the paragraph.

9.	List-I	List-II
	(I) Condensation at normal melting point ; 1 atm	(P) $\Delta G < 0$
	(II) Fusion at normal boiling point ; 1 atm	(Q) $\Delta G = 0$
	(III) Vaporisation at triple point	(R) $\Delta S_{\text{system}} > 0$
	(IV) Freezing at normal melting point ; 1 atm	(S) $\Delta S_{\text{system}} < 0$
		(T) $ \Delta S_{\text{system}} > \Delta S_{\text{surrounding}} $
		(U) $ \Delta S_{\text{system}} < \Delta S_{\text{surrounding}} $

Which is correctly matched for an endothermic process ?

- (A) (I) $\rightarrow$ P, S, T (B) (II) $\rightarrow$ P, R, T
 (C) (II) $\rightarrow$ P, R, U (D) (III) $\rightarrow$ Q, R, T

Space for Rough Work

10.

List-I

List-II

(I) Condensation at normal melting point ; 1 atm

 (P) $\Delta G < 0$

(II) Fusion at normal boiling point ; 1 atm

 (Q) $\Delta G = 0$

(III) Vaporisation at triple point

 (R) $\Delta S_{\text{system}} > 0$

(IV) Freezing at normal melting point ; 1 atm

 (S) $\Delta S_{\text{system}} < 0$

 (T) $|\Delta S_{\text{system}}| > |\Delta S_{\text{surrounding}}|$

 (U) $|\Delta S_{\text{system}}| < |\Delta S_{\text{surrounding}}|$

Which is correctly matched for an exothermic process ?

 (A) (I) $\rightarrow$ Q, S

 (B) (I) $\rightarrow$ Q, S, U

 (C) (III) $\rightarrow$ Q, R

 (D) (IV) $\rightarrow$ Q, S

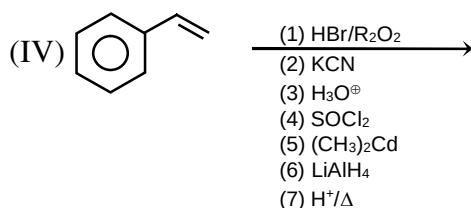
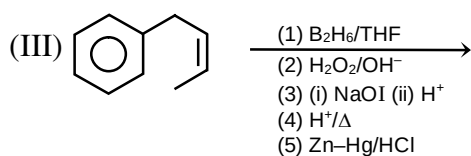
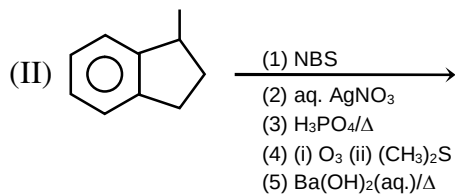
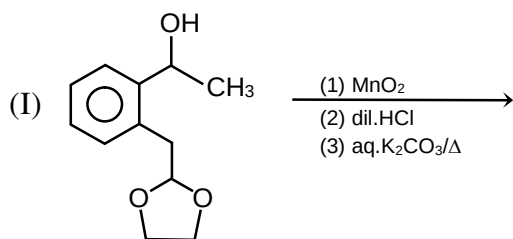
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Answer the following questions by appropriately matching the list based on the information given in the paragraph.

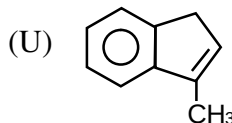
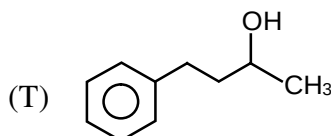
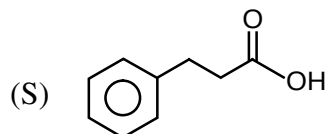
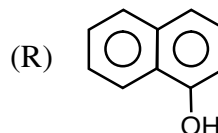
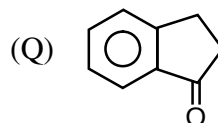
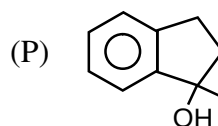
List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I.

11.

List-I



List-II



Which of the following options has correct combination considering List-I and List-II?

(A) (III), (S), (R), (T)

(B) (IV), (Q), (R), (S)

(C) (III), (T), (Q), (S)

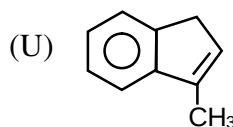
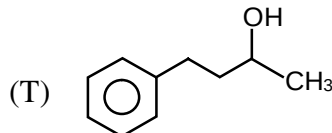
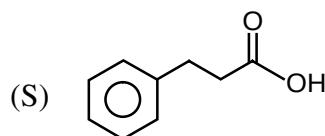
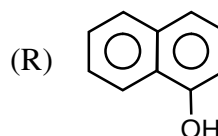
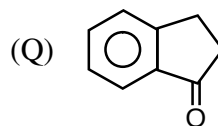
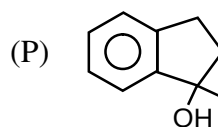
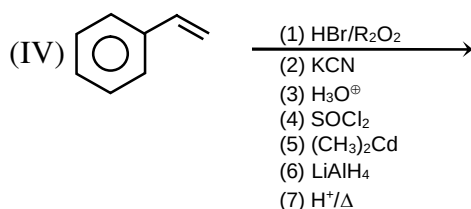
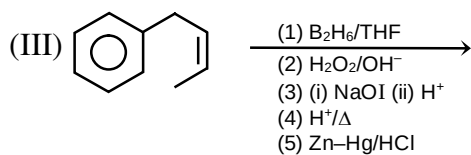
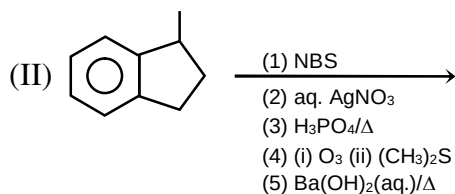
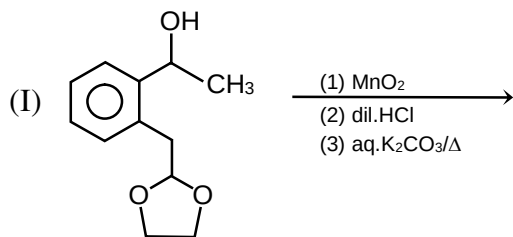
(D) (IV), (Q), (U)

Space for Rough Work

12.

List-I

List-II



Which of the following options has correct combination considering List-I and List-II?

(A) (I), (Q), (T), (U)

(B) (II), (P), (R), (U)

(C) (II), (P), (S), (T)

(D) (I), (S), (Q), (R)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. In order to calculate the ionic product of water accurately, the following galvanic cell is used.



0.1M 0.01M

By measuring cell potential at appropriate concentration the value of ionic product can be calculated. Assuming KOH and KCl to be completely dissociated and $E_{\text{Cl}^-(\text{aq})|\text{AgCl(s)}|\text{Ag}}^0 = 0.3 \text{ V}$

If E_{cell} at 320 K is 1.02 V, Determine the value of $K_w \times 10^{14}$ at 320 K.

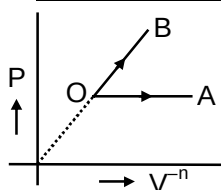
$$\left(\text{Give : } \frac{2.303 \times R \times 320}{F} = 0.06 \right)$$

2. How many of the following can oxidise SO_2 under appropriate conditions?

Acidified KMnO_4 , Acidified $\text{K}_2\text{Cr}_2\text{O}_7$, aqueous FeCl_3 , Cl_2 gas with charcoal, Heat with O_2 and V_2O_5 , O_3 , PCl_5 , Br_2 + water, KIO_3 + H^+

Space for Rough Work

3.

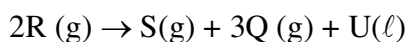


A sample of non linear polyatomic ideal gas is used for process OA & OB if $T_A = T_B$ and

$$\frac{q_{OA}}{q_{OB}} = 2$$

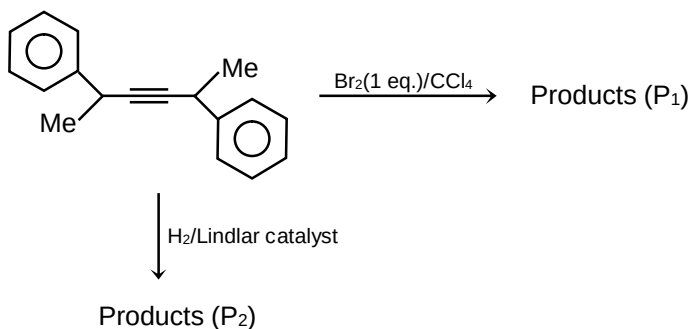
Determine n.

4. Decomposition of a compound 'R' at temperature T, according to the following reaction is a second order reaction



Starting with only R(g) at 800 mm pressure, the pressure at the end of 1 hour is found to increase by 80%. Determine the partial pressure of S(g) at the end of 5 hours (given vapour pressure of U(l) at this temperature is 40 mm & vapour-liquid equilibrium of U is established in less than 1 hour)

5.

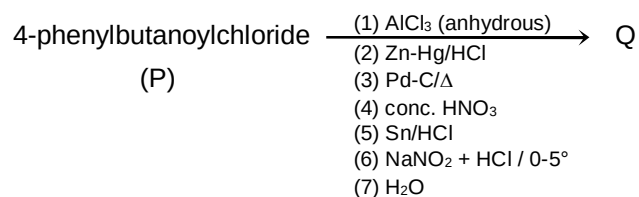


Total number of chiral products in P_1 and P_2 is :

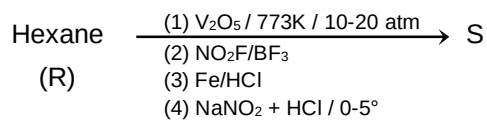
Space for Rough Work

6. Scheme 1 and 2 describe the conversion of **P** to **Q** and **R** to **S**, respectively. Scheme 3 describes the synthesis of **T** from **Q** and **S**. The total number of sp^2 hybridised atoms in **T** is:

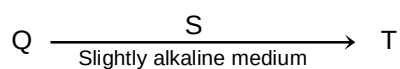
Scheme 1:



Scheme 2:



Scheme 3:



Space for Rough Work

PART-3 : MATHEMATICS

SECTION- I(i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 marks;
 choosing **ONLY** (B) will get +1 marks;
 choosing **ONLY** (D) will get +1 marks;
 choosing no option (i.e. the question is unanswered) will get 0 marks, and
 choosing any other combination of options will get -1 mark.

1. let $(1 - x^3)^n = \sum_{r=0}^n a_r x^r (1-x)^{3n-2r} \forall x \in \mathbb{R} - \{1\}$ then which of the following statement is/are correct ?
- (A) If $n = 7$ then maximum value of a_r is at $r = 5$ or 6
- (B) $\sum_{r=0}^n a_r$ is divisible by 2^n
- (C) $\frac{a_3}{a_2}$ is integer $\forall n \in \mathbb{N}$
- (D) $\sum_{r=0}^n a_r$ is not divisible by 2^n

Space for Rough Work

2. Let A and B are square matrix of odd order such that $A^T = A^2 - B$ and $B^T = B^2 - A$. Then which of the following statement is/are correct?
- (A) $|B^2 - A^2| = 0$ (B) $|A| \neq |B|$
(C) $|A| = |B|$ (D) $|B^2 - A^2| \neq 0$
3. Let position vectors of points A, B, C are $\vec{a}$, $\vec{b}$, $\vec{c}$ respectively where $\vec{c} = x\vec{a} + y\vec{b}$
- (A) If $x + y = 1$, $x > 0$, $y < 0$ then C lies on line AB but not on line segment AB
(B) If $x > 0$, $y > 0$, $x + y < 1$ then C lies inside the triangle ΔABO
(C) If $x < 0$, $y > 0$ and $x + y > 1$ then on plane containing points O, A, B, the points C and origin lies on opposite side of line AB.
(D) If $x \leq 0$, $y \leq 0$ and $x + y = -1$ then locus of point C is reflection of line segment AB about origin

Space for Rough Work

4. Suppose that $f(x)$ is a differentiable invertible function with $f'(x) \neq 0$ and $h(x) = \int_1^x f(t) dt$. Given that $f(1) = f'(1) = 1$ and $g(x)$ is inverse of $f(x)$. Let $G(x) = x^2 g(x) - xh(g(x)) \forall x \in \mathbb{R}$. Which of the following are correct-
- (A) $G'(1) = 2$ (B) $G'(1) = 3$ (C) $G''(1) = 2$ (D) $G''(1) = 3$
5. $\left(\int_0^1 x^{101} (A + Bx^{10}) \sqrt{1+x^{10}} dx \right)^2 = C$ then-
- (A) $A = 68$ (B) $B = 78$ (C) $9C - 32 = 0$ (D) $9C + 22 = 0$

Space for Rough Work

6. Let $p(x)$ & $q(x)$ are monic quadratics such that $p(x) = -q(100 - x)$ and graph of $p(x)$ contains the vertex of $q(x)$. If roots of $p(x)$ and $q(x)$ are α, β and γ, δ respectively ($\alpha > \beta > \gamma > \delta$) such that $|(\gamma - \beta)| = 150$ and $|\delta - \alpha| = m + n\sqrt{p}$ then -
 (A) $m = 450$ (B) $n = 300$ (C) $m + p = 452$ (D) $n + p = 305$
7. If the function $y = f(x)$ is satisfying the condition $f(x) f'(x) f''(x) < 0 \forall x \in \mathbb{R}$, then which of the following statement is/are true?
 (A) $f(x)$ is decreasing function
 (B) $f(x)$ and $f''(x)$ are of opposite sign
 (C) If $f(x)$ cuts the y -axis at $(0, 1)$ then $f(f(x))$ lies between $(0, 1)$
 (D) curve of $y = f^{-1}(x)$ does not intersect y -axis
8. Let $(x + 1)(x + 2) \dots (x + n) = A_0 + A_1x + A_2x^2 + \dots + A_nx^n$, then-
 (A) $A_0 - 2A_1 + 2^2A_2 - \dots + (-2)^n A_n = 0$
 (B) $A_0 + 2A_1 + 3A_2 + \dots + (n + 1)A_n = (n + 1)! \left(1 + \frac{1}{2} + \dots + \frac{1}{n+1} \right)$
 (C) $nA_0 + (n - 1)A_1 + \dots + A_{n-1} = (n + 1)! \left(\frac{1}{2} + \frac{2}{3} + \dots + \frac{n}{n+1} \right)$
 (D) none of these

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **Two (02)** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

Answer the question number 9 and 10 by appropriately matching the lists based on the information given in the paragraph

Let combined equation of L_1, L_2, L_3, L_4 is $ax^4 + bx^3y + cx^2y^2 + dxy^3 + ay^4 = 0$. If the lines L_1, L_2 bisects the angle between the L_3, L_4 and vice versa, then $\frac{c}{a} = k_1, \frac{d}{b} = k_2$. If L_1, L_2 are parallel to L_3, L_4 respectively but L_1 is not perpendicular to L_2 then $\frac{d^2 - 4ac}{4a^2} = k_3, \frac{4ac - 8a^2}{d^2} = k_4$.

List-I

- (I) k_1 is equal to
- (II) k_2 is equal to
- (III) k_3 is equal to
- (IV) k_4 is equal to

List-II

- (P) -1
- (Q) 1
- (R) -2
- (S) 3
- (T) -6
- (U) 5

9. Which of the following is the only **CORRECT** combination?
 (A) (II), (Q) (B) (I), (U) (C) (II), (R) (D) (I), (T)
10. Which of the following is the only **INCORRECT** combination?
 (A) (III), (R) (B) (IV), (Q) (C) (III), (S) (D) (II), (P)

Space for Rough Work

Answer the question number 11 and 12 by appropriately matching the lists based on the information given in the paragraph

Let two term of H.P. are 12 and 15. All terms of this H.P. are natural numbers. If among the largest possible such H.P. (i.e., is number of terms in H.P. are maximum), the three of the terms are a, b, c which are in increasing G.P. then

List-I

- (I) one of the term of such H.P. is
- (II) a is less than
- (III) c is greater than
- (IV) natural multiples of b can be

List-II

- (P) 9
- (Q) 10
- (R) 18
- (S) 20
- (T) 30
- (U) 60

11. Which of the following is the only CORRECT combination?

- (A) (II), (P) (B) (III), (U) (C) (II), (Q) (D) (III), (R)

12. Which of the following is the only INCORRECT combination?

- (A) (I), (S) (B) (IV), (T) (C) (IV), (R) (D) (I), (T)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. If in a 4×4 matrix the sum of elements of each row, column and both the main diagonals is 1, then sum of the four corner elements is equal to-
2. Let $f(x) = 2x(2 - x)$, $0 \leq x \leq 2$. If the number of roots and sum of all roots of equation $f(f(f(x))) = \frac{x}{2}$ are equal to 'a' and 'b' respectively, then $(b + a)$ is equal to-
3. If largest term in the sequence $\frac{1}{201}, \frac{4}{208}, \frac{9}{227}, \frac{16}{264}, \dots$ is n^{th} term, then value of $\frac{1}{T_{n-3}}$ is equal to-

Space for Rough Work

4. On a semicircle with diameter AB and centre O point E and C are marked in such a way that OE is perpendicular to AB, and the chord AC intersects the segment OE at a point D which is interior to the semicircle. If $\angle CBA = \frac{\pi}{k}$ such that a circle can be inscribed into the quadrilateral OBCD then value of k is equal to-
5. A straight line PQ touches the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the circle $x^2 + y^2 = r^2$ ($3 < r < 4$). RS is a focal chord of the ellipse which is parallel to PQ and meets the circle at point R and S. Then the length of RS is equal to-
6. Find sum of all integral values of r such that all roots of the polynomial $f(x) = x^3 - (r-1)x^2 - 11x + 4r$ are also integer

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work